

Date _____

Question 06:

Pseudocode:

START

DECLARE current SOC, Req. SOC, duration, Time, Req. Charge, Charge Cost, Park Cost, Total Discount, Total Payable, Base Rate, Base Park, Discount P, Rate Charge Cost, Charge Discount, Park Discount: FLOAT DECLARE eligible: BOOLEAN

DECLARE vehicle, member, disabled, available: CHAR

DECLARE Priority, Time-Status, Message: STRING

OUTPUT "Enter vehicle type (E/H): "

INPUT Vtype

OUTPUT "Enter Battery charge level as a percentage: "

INPUT current SOC

OUTPUT "Enter Required charging level as a percentage: "

INPUT Req. SOC

OUTPUT "Enter expected parking duration in hours: "

INPUT duration

OUTPUT "Enter Current time in 24-hour format: "

INPUT Time

OUTPUT "Do you have a parking membership (Y/N): "

INPUT member

OUTPUT "Do you have disabled-person priority status? (Y/N): "

INPUT disabled

OUTPUT "Is the charging station currently available? (Y/N): "

INPUT available

chargeC ← 0

ParkC ← 0

Total_discount ← 0

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```
Req_charge ← 0
Priority ← "None"
Time_Status ← "None"
Message ← "", eligible ← TRUE, Req_charge ← Req_SOC - current_SOC
IF available = 'N' THEN
    IF vehicle = 'H' THEN
        OUTPUT "Charging unavailable; Parking only."
    ELSE
        OUTPUT "No charging slots available."
    ENDIF
    eligible ← FALSE
ELSE IF vehicle = 'H' AND current_SOC ≥ 40 THEN
    OUTPUT "Vehicle does not qualify for EV charging."
    eligible ← FALSE
ELSE IF Req_charge ≤ 0 THEN
    OUTPUT "No charging required"
    eligible ← FALSE
ENDIF
IF eligible = TRUE THEN
    IF current_SOC ≤ 15 AND Req_SOC ≥ 80 THEN
        Priority ← "Emergency Charging Priority"
    ELSE IF disabled = 'X' OR (member = 'Y' AND current ≤ 30) THEN
        Priority ← "Priority Charging"
    ELSE
        Priority ← "Normal Charging"
    ENDIF
    Discount_P ← 0.0
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IF Time ≥ 17 AND Time < 22 THEN

Time_status $\leftarrow$ "Peak"

Base_Rate $\leftarrow 50$

IF member = 'Y' AND Priority $\neq$ "Emergency Charging Priority" THEN

Discount_P $\leftarrow 0.10$

ENDIF

ELSE

Time_status $\leftarrow$ "Off-peak"

Base_Rate $\leftarrow 35$

IF member = 'Y' AND Priority $\neq$ "Emergency Charging Priority" THEN

Discount_P $\leftarrow 0.20$

ENDIF

ENDIF

Raw_Charge_Cost $\leftarrow$ Req_charge * Base_Rate

Charge_Discount $\leftarrow$ Raw_Charge_Cost * Discount_P

Charge_Cost $\leftarrow$ Raw_Charge_Cost - Charge_Discount

Total_Discount $\leftarrow$ Total_Discount +

ENDIF

IF duration ≤ 2 THEN

Base_Park = 200

ELSEIF duration ≤ 5 THEN

Base_Park = 400

ELSE

Base_Park = 700

ENDIF

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```
IF disabled = 'Y' THEN
    Park_Cost = 0
    Total_Discount ← Total_Discount + Base_Park
ELSE IF member = 'Y' THEN
    Park_Discount ← Base_Park * 0.20
    Park_Cost ← Base_Park - Park_Discount
    Total_Discount ← Total_Discount + Park_Discount
ELSE
    Park_Cost ← Base_Park
ENDIF
IF duration > 8 THEN
    Message ← "Long stay warning... Please relocate your vehicle after charging."
ELSE
    Message ← "Standard parking duration"
ENDIF
Total_Payable ← Charge_Cost + Park_Cost

OUTPUT "Vehicle type:" vehicle
OUTPUT "Current Battery:" current_SoR " | Required Battery:" Req_SoR
OUTPUT "Charging priority:" Priority
OUTPUT "Time status:" Time_Status
OUTPUT "Charging Cost:" Charge_Cost
OUTPUT "Parking Cost:" Park_Cost
OUTPUT "Total Discounts applied:" Total_Discount
OUTPUT "Final amount:" Total_Payable
OUTPUT Message
END
```

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IPD

INPUT	Process	OUTPUT
- v type (Vehicle type: (E/H)) - current SOC (current battery level %) - Req. SOC (Required battery level %) - duration (Expected parking duration in hours) - Time (current time in 24-hour format) - member (Parking membership: Y/N) - disabled (Disabled-person priority: Y/N) - available (charging station availability: Y/N)	(1) Eligibility & Setup - calculate Req. Charge ($\text{Req. SOC} - \text{current SOC}$) - Check if charging is available, if vehicle qualifies (Hybrid with $\geq 40\%$ SOC don't) and if charge is needed ($\text{Req. Charge} > 0$). Set eligible flag (2) Determine Priority (Emergency, Priority, or Normal) based on, current SOC, Req. SOC, disabled, and member status. (3) Determine Time Status (Peak or Off-Peak) based on Time (Peak is 17 to < 22). Set base Rate (\$5 for peak, \$3 for off-peak). (4) Calculate Raw Charge Cost ($\text{Req. Charge} * \text{Base Rate}$) Apply Discount-P based on member and priority status to calculate Charge - Cost - Determine Base Park fee - Apply parking discounts,	Status Messages - "Charging unavailable" - "No charging slots available" - Vehicle does not qualify for EV charging - No charging required Final Receipt Outputs - Vehicle type - Current Battery & Required Battery - Time status - Charging Cost - Parking cost - Total discounts applied - Final amount = - Total Payable - Message (Long stay warning or direct)

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PAC

Data Given	Processing required	Required Results
- v type (Vehicle type: (E/H)) - current SOC (current battery level %) - Req. SOC (Required battery level %) - duration (Expected parking duration in hours) - Time (current time in 24-hour format) - member (Parking membership: Y/N) - disabled (Disabled-person priority: Y/N) - available (charging station availability: Y/N)	(1) Eligibility & Setup - calculate Req. Charge ($\text{Req. SOC} - \text{current SOC}$) - Check if charging is available, if vehicle qualifies (Hybrid with $\geq 40\%$ SOC don't) and if charge is needed ($\text{Req. Charge} > 0$). Set eligible flag. (2) Determine Priority (Emergency, Priority, or Normal) based on, current SOC, Req. SOC, disabled, and member status. (3) Determine Time Status (Peak or Off-Peak) based on Time (Peak is 17 to <22). Set base Rate (\$5 for peak, \$3 for Off-peak). (4) Calculate Raw Charge Cost ($\text{Req. Charge} * \text{Base Rate}$) - Apply Discount P based on member and priority status to calculate Charge - Cost - Determine Base Park fee - Apply parking discounts,	Status Messages - "Charging unavailable" - "No charging slots available" - Vehicle don not qualify for EV charging - No charging required Final Receipt Outputs - Vehicle type - Current Battery & Required Battery - Time status - Charging cost - Parking cost - Total discounts applied - Final amount - Total Payable - Message (long stay warning or closed)

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Question 06

START



OUTPUT "Enter vehicle type (E/H):"

INPUT Vtype

OUTPUT "Enter battery charge level:"

INPUT current SOC

OUTPUT "Enter required charging level:"

INPUT Req-SOC

OUTPUT "Enter expected duration:"

INPUT duration

OUTPUT "Enter current time in 24-hour format:"

INPUT Time

OUTPUT "Parking membership? (Y/N)"

INPUT member

OUTPUT "Do you have disabled person priority?"

INPUT disabled

OUTPUT "Is the charging station available?"

INPUT available



Charge_Cost ← 0

Park_Cost ← 0

Total_Discount ← 0

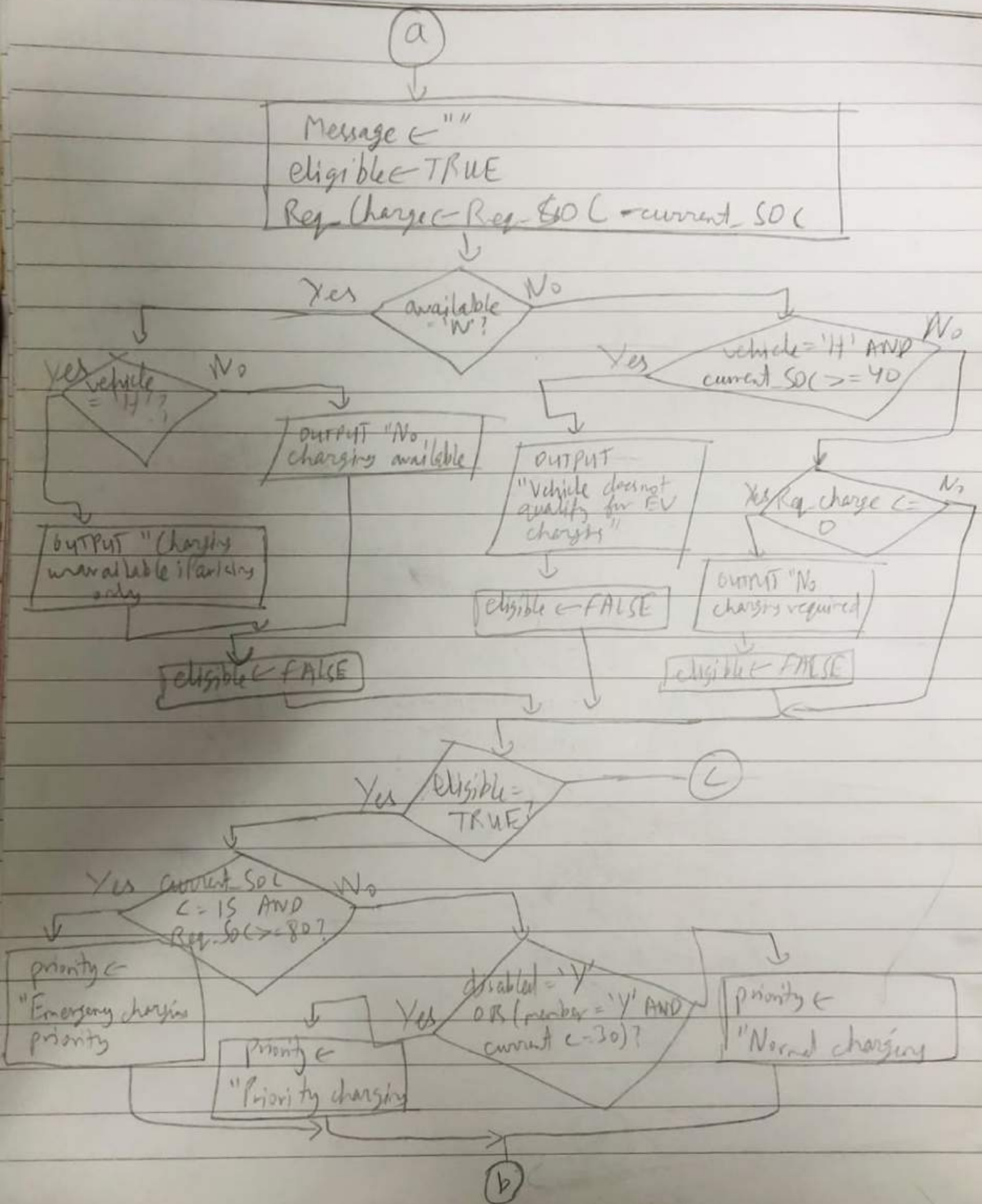
Req_charge ← 0

Priority ← "None"

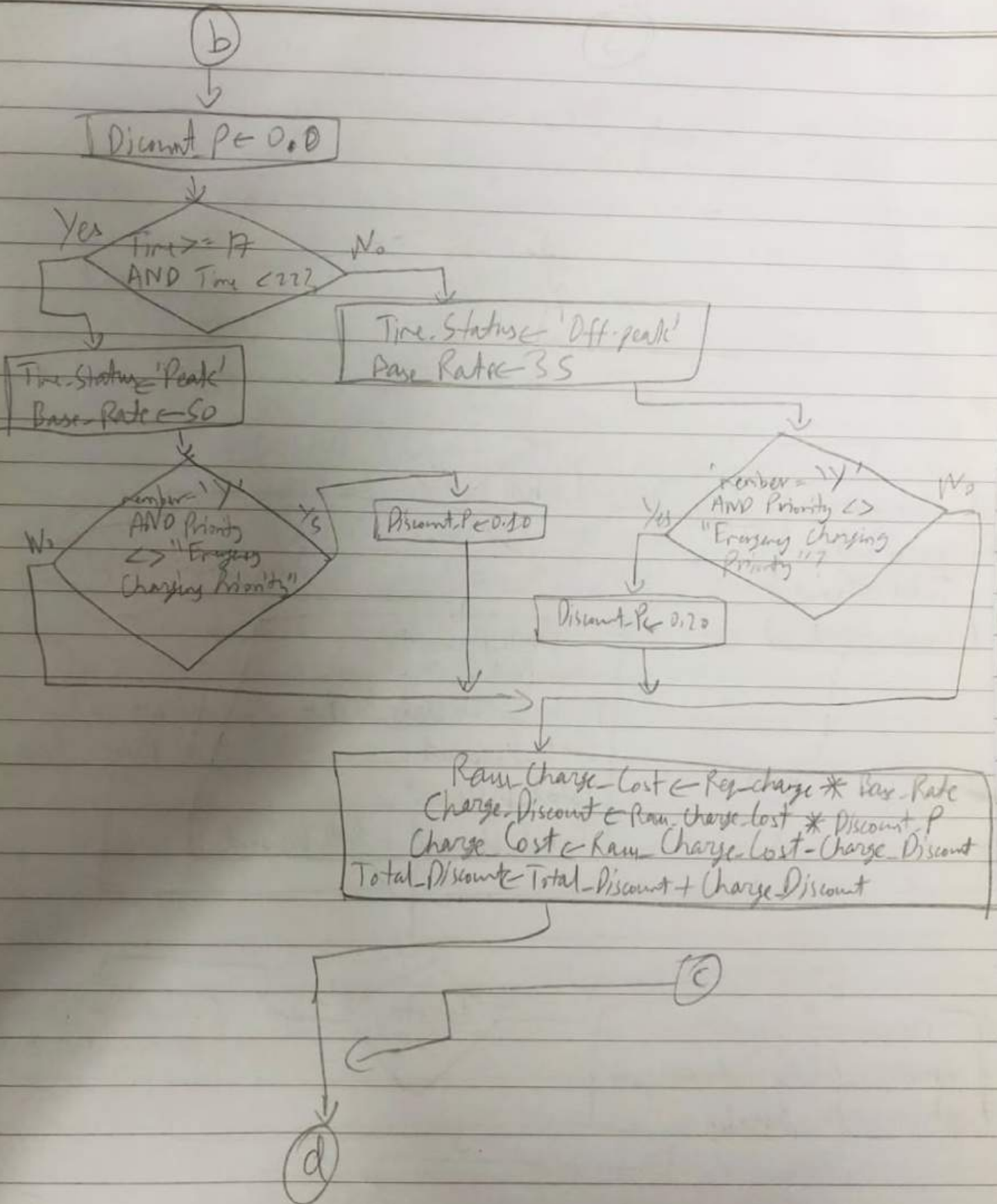
Time_Status ← "None"



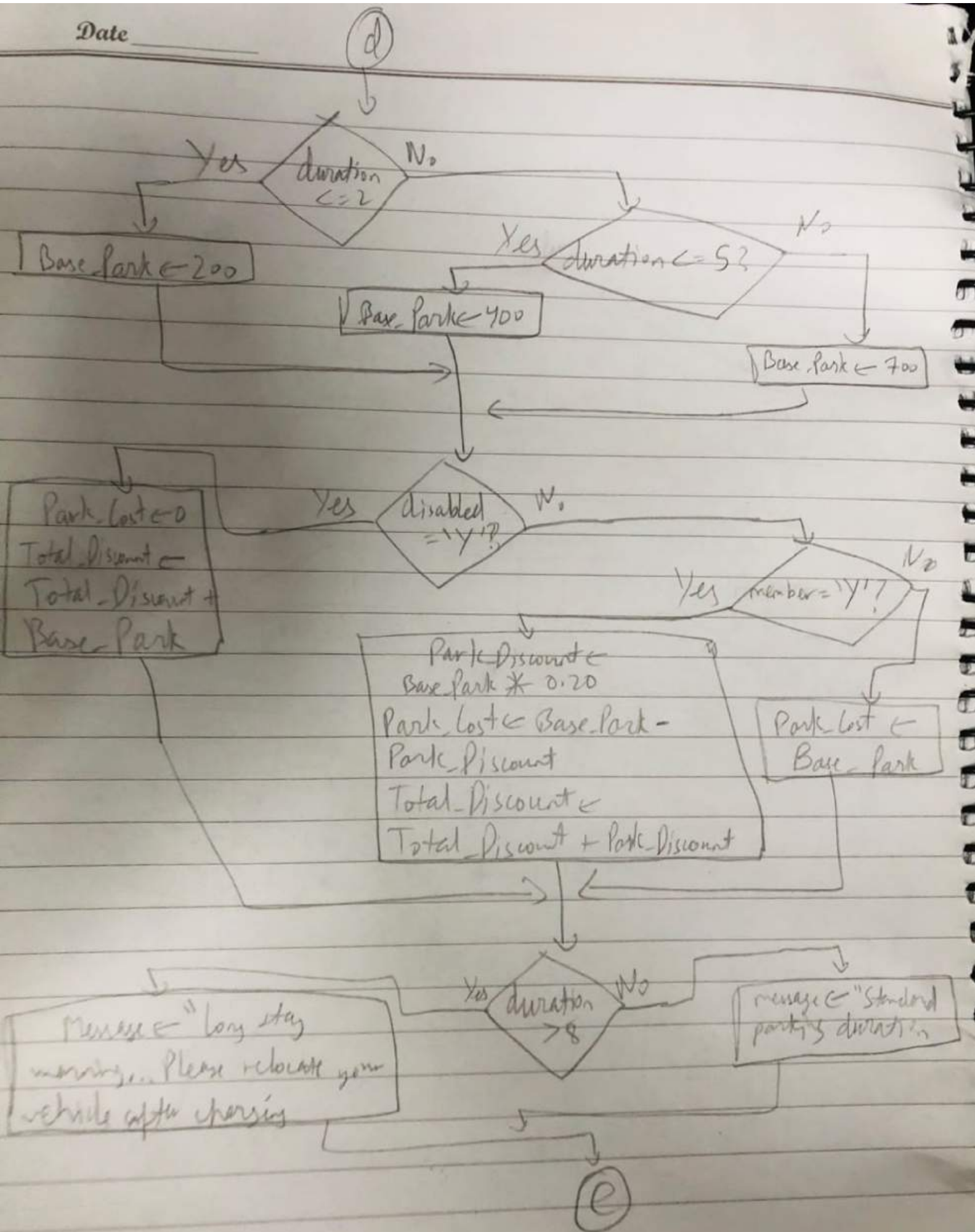
a



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(e)



Total Payable = Charge_Lost + Park_Lost



OUTPUT "Vehicle type:" vehicle
OUTPUT "Current Battery:" current_SOC | Required
Battery:" Req SOC
OUTPUT "Charging priority:" priority
OUTPUT "Time status:" Time_Status
OUTPUT "Charging cost:" Charge_Lost
OUTPUT "Parking cost:" Park_Lost
OUTPUT "Total Discounts applied:" Total_Discount
OUTPUT "Final amount:" Total_Payable
OUTPUT Message



END